In [1]:
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# Q2 (a)
```

In [2]:
```
import numpy as np
from scipy.interpolate import lagrange

# Given data: time (t) and altitude (h)
t = np.array([0, 2, 5, 9, 15])
h = np.array([0.0, 12.1, 65.5, 197.6, 499.0])

# Create Lagrange interpolating polynomial
poly = lagrange(t, h)

# Print the polynomial
print("Lagrange interpolating polynomial:")
print(poly)
```

```
Lagrange interpolating polynomial:
           4           3         2
-0.001033 x - 0.00292 x + 2.411 x + 1.249 x
```

In [3]:
```
# Q2 (b)
```

In [4]:
```
import numpy as np
from scipy.interpolate import lagrange

# Given data
t = np.array([0, 2, 5, 9, 15])
h = np.array([0.0, 12.1, 65.5, 197.6, 499.0])

# Lagrange interpolation
poly = lagrange(t, h)

# Get polynomial coefficients (highest degree first)
coeffs = poly.coefficients

# First derivative (velocity)
velocity_coeffs = np.polyder(coeffs)

# Second derivative (acceleration)
acceleration_coeffs = np.polyder(velocity_coeffs)

# Evaluate at t = 7
t_val = 7
velocity = np.polyval(velocity_coeffs, t_val)
acceleration = np.polyval(acceleration_coeffs, t_val)

# Print results
print("Velocity at t = 7 seconds:", round(velocity, 2), "m/s")
print("Acceleration at t = 7 seconds:", round(acceleration, 2), "m/s²")
```

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Velocity at t = 7 seconds: 33.15 m/s
Acceleration at t = 7 seconds: 4.09 m/s²
```

In [ ]:

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